KAIZEN MATHS

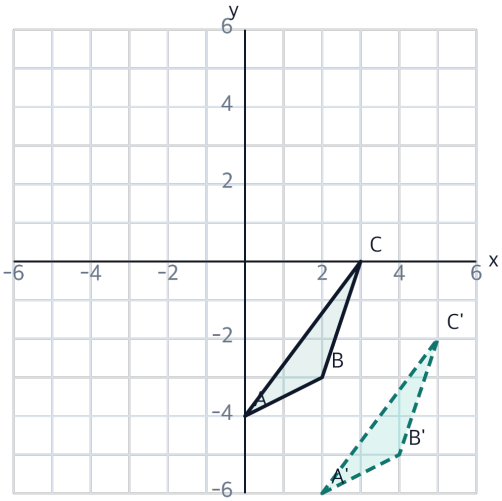
Lesson starter

Transformations

Describe Transformations

Name _____

SA1



Describe fully the single transformation that maps shape ABC onto shape A'B'C'.

GEO-C8E6-L3 · SA

Show your working

KAIZEN MATHS

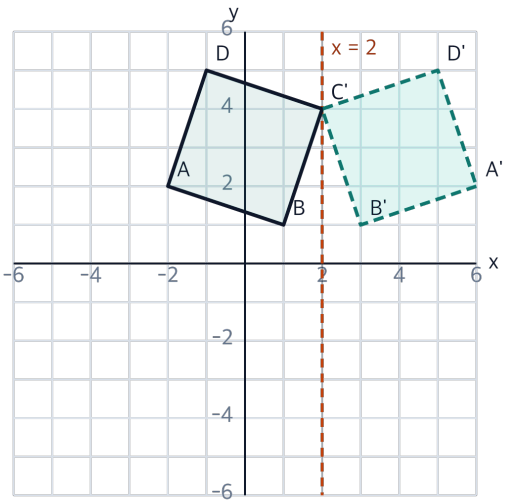
Lesson starter

Transformations

Describe Transformations

Name _____

SB1



Describe fully the single transformation that maps shape ABCD onto shape A'B'C'D'.

GEO-C8E6-L3 · SB

Show your working

KAIZEN MATHS

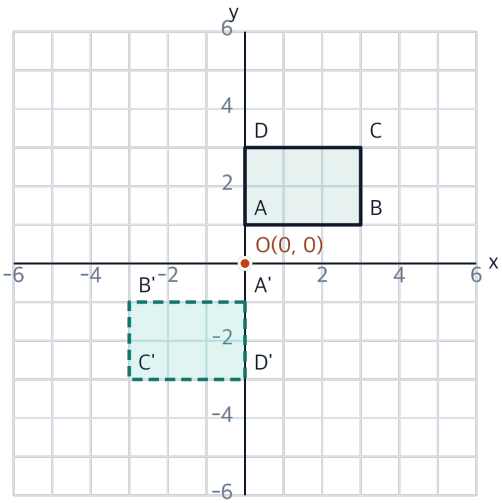
Lesson starter

Transformations

Describe Transformations

Name _____

SC1



Describe fully the single transformation that maps shape ABCD onto shape A'B'C'D'.

GEO-C8E6-L3 · SC

Show your working

KAIZEN MATHS

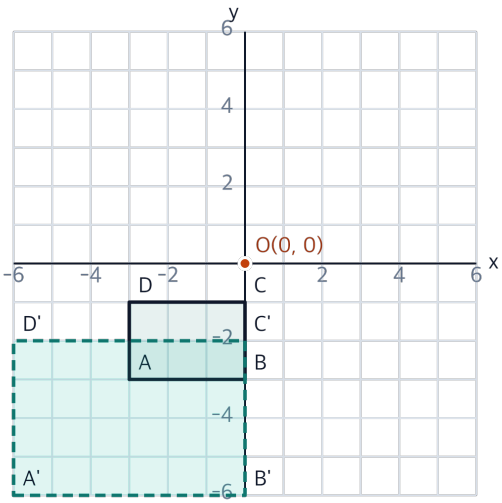
Lesson starter

Transformations

Describe Transformations

Name _____

SD1



Describe fully the single transformation that maps shape ABCD onto shape A'B'C'D'.

GEO-C8E6-L3 · SD

Show your working

KAIZEN MATHS

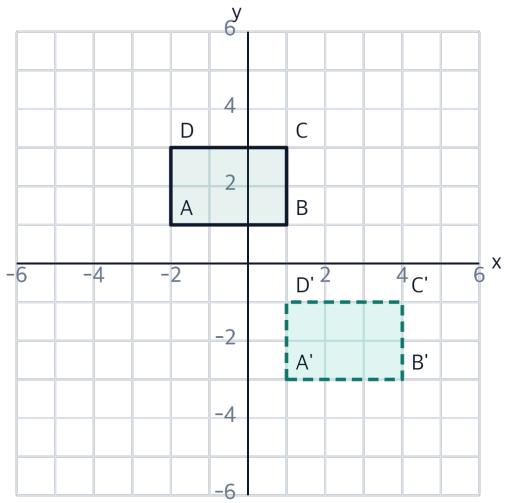
Exit ticket

Transformations

Describe Transformations

Name _____

EA1



Describe fully the single transformation that maps shape ABCD onto shape A'B'C'D'.

GEO-C8E6-L3 · EA

Show your working

KAIZEN MATHS

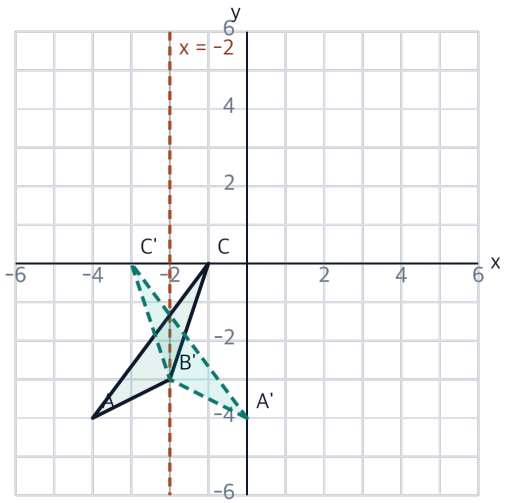
Exit ticket

Transformations

Describe Transformations

Name _____

EB1



Describe fully the single transformation that maps shape ABC onto shape A'B'C'.

GEO-C8E6-L3 · EB

Show your working

KAIZEN MATHS

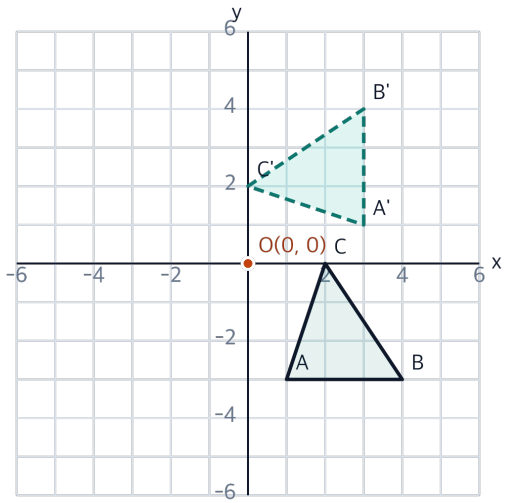
Exit ticket

Transformations

Describe Transformations

Name _____

EC1



Describe fully the single transformation that maps shape ABC onto shape A'B'C'.

GEO-C8E6-L3 · EC

Show your working

KAIZEN MATHS

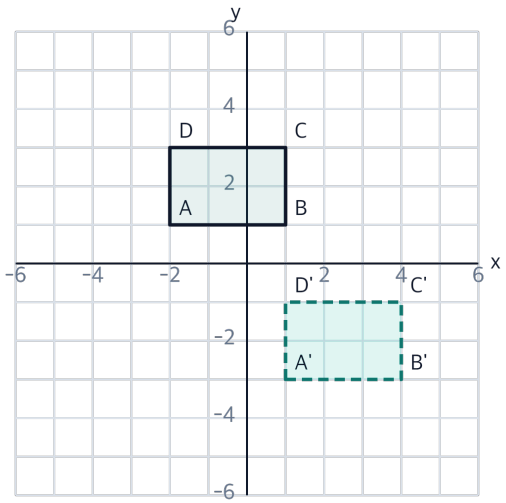
Exit ticket

Transformations

Describe Transformations

Name _____

EA1



Describe fully the single transformation that maps shape ABCD onto shape A'B'C'D'.

GEO-C8E6-L3 · EA

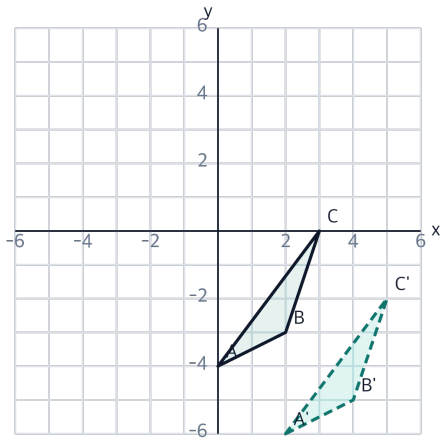
Show your working

Teacher answers & working

Transformations · Describe Transformations

GEO-C8E6-L3 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

SA1



Describe fully the single transformation that maps shape ABC onto shape A'B'C'.

Answer:

Translation by vector $\begin{pmatrix} 2 \\ -2 \end{pmatrix}$

Working:

Compare matching vertices, for example A to A'.

Horizontal change = 2 and vertical change = -2.

The same movement maps every vertex onto its image.

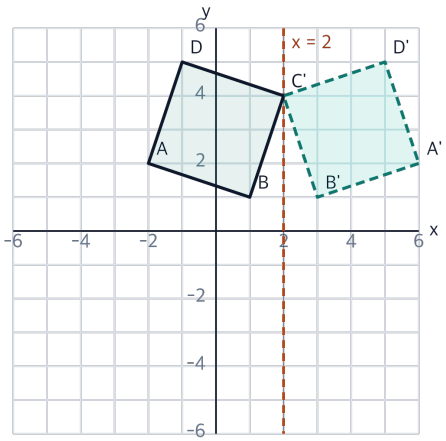
Transformation: translation by vector $\begin{pmatrix} 2 \\ -2 \end{pmatrix}$.

Teacher answers & working

Transformations · Describe Transformations

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SB1



Describe fully the single transformation that maps shape ABCD onto shape A'B'C'D'.

Answer:

Reflection in the line x = 2

Working:

Join each vertex to its image and look for the perpendicular bisector.

The same mirror line lies halfway between every matching pair of vertices.

The mirror line is x = 2.

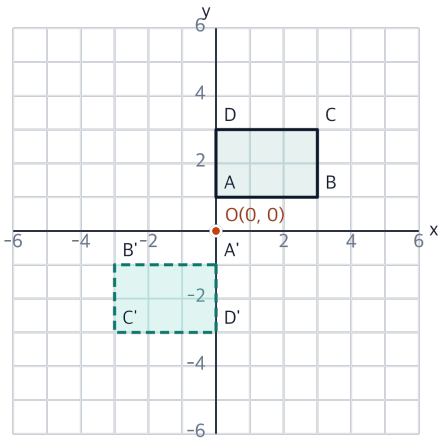
Transformation: reflection in the line x = 2.

Teacher answers & working

Transformations · Describe Transformations

GEO-C8E6-L3 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

SC1



Describe fully the single transformation that maps shape ABCD onto shape A'B'C'D'.

Answer:

Rotation 180 degrees about the origin

Working:

The object and image stay the same distance from the centre of rotation.

The centre is the origin (0, 0).

The direction and angle from each vertex to its image show a 180 degrees turn.

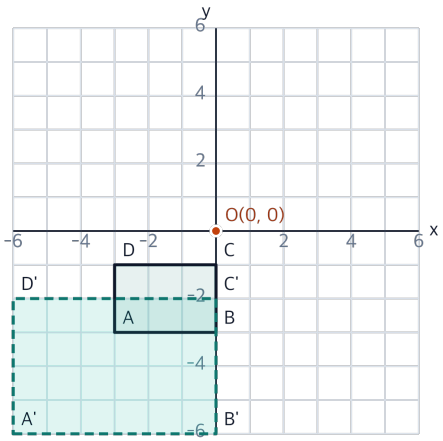
Transformation: rotation 180 degrees about the origin.

Teacher answers & working

Transformations · Describe Transformations

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SD1



Describe fully the single transformation that maps shape ABCD onto shape A'B'C'D'.

Answer:

Enlargement with scale factor 2 from centre (0, 0)

Working:

The object and image lie on rays from the centre of enlargement.

The centre is the origin (0, 0).

Compare a matching distance from the centre: image distance = 2 times object distance.

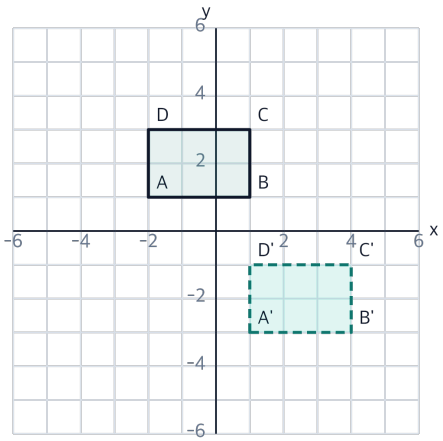
Transformation: enlargement with scale factor 2 from centre (0, 0).

Teacher answers & working

Transformations · Describe Transformations

GEO-C8E6-L3 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

EA1



Describe fully the single transformation that maps shape ABCD onto shape A'B'C'D'.

Answer:

Translation by vector $\begin{pmatrix} 3 \\ -4 \end{pmatrix}$

Working:

Compare matching vertices, for example A to A'.

Horizontal change = 3 and vertical change = -4.

The same movement maps every vertex onto its image.

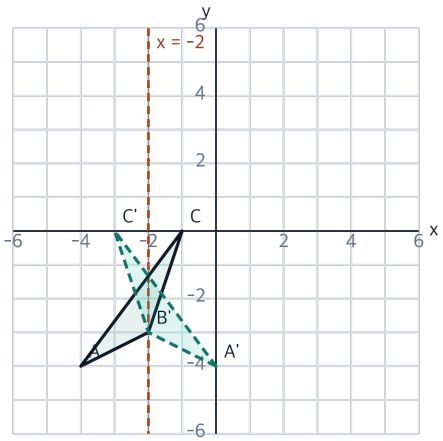
Transformation: translation by vector $\begin{pmatrix} 3 \\ -4 \end{pmatrix}$.

Teacher answers & working

Transformations · Describe Transformations

GEO-C8E6-L3 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

EB1



Describe fully the single transformation that maps shape ABC onto shape A'B'C'.

Answer:

Reflection in the line x = -2

Working:

Join each vertex to its image and look for the perpendicular bisector.

The same mirror line lies halfway between every matching pair of vertices.

The mirror line is x = -2.

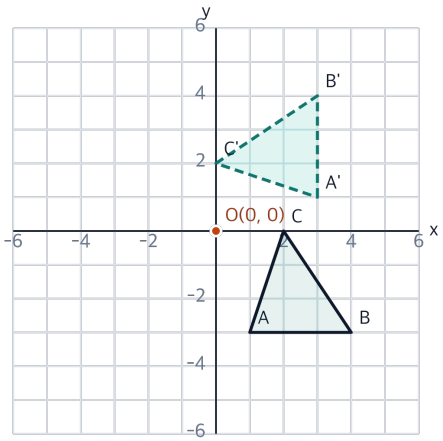
Transformation: reflection in the line x = -2.

Teacher answers & working

Transformations · Describe Transformations

GEO-C8E6-L3 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

EC1



Describe fully the single transformation that maps shape ABC onto shape A'B'C'.

Answer:

Rotation 90 degrees anticlockwise about the origin

Working:

The object and image stay the same distance from the centre of rotation.

The centre is the origin (0, 0).

The direction and angle from each vertex to its image show a 90 degrees anticlockwise turn.

Transformation: rotation 90 degrees anticlockwise about the origin.